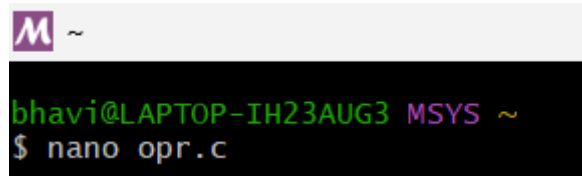


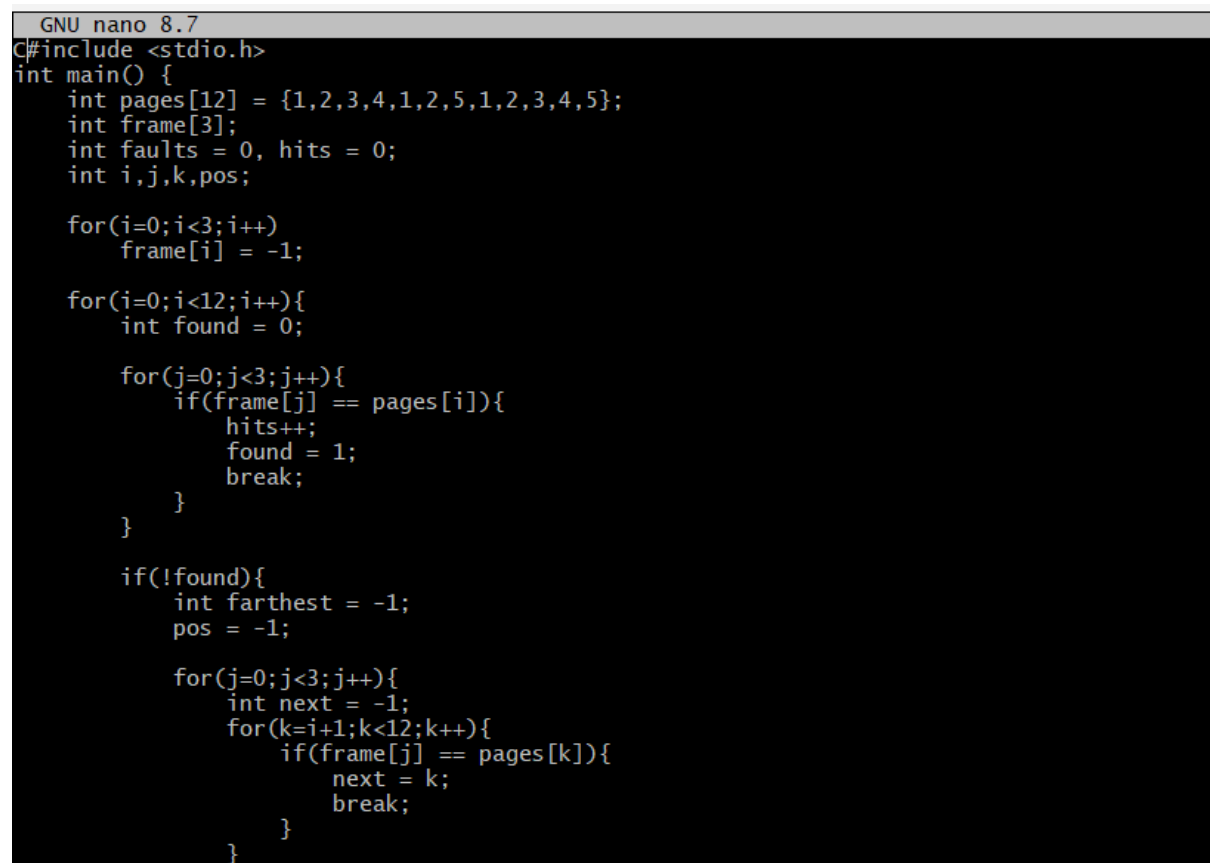
Practical No : 09

Aim : A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Input :



```
M ~  
bhavi@LAPTOP-IH23AUG3 MSYS ~  
$ nano opr.c
```



```
GNU nano 8.7  
C#include <stdio.h>  
int main() {  
    int pages[12] = {1,2,3,4,1,2,5,1,2,3,4,5};  
    int frame[3];  
    int faults = 0, hits = 0;  
    int i,j,k,pos;  
  
    for(i=0;i<3;i++){  
        frame[i] = -1;  
    }  
  
    for(i=0;i<12;i++){  
        int found = 0;  
  
        for(j=0;j<3;j++){  
            if(frame[j] == pages[i]){  
                hits++;  
                found = 1;  
                break;  
            }  
        }  
  
        if(!found){  
            int farthest = -1;  
            pos = -1;  
  
            for(j=0;j<3;j++){  
                int next = -1;  
                for(k=i+1;k<12;k++){  
                    if(frame[j] == pages[k]){  
                        next = k;  
                        break;  
                    }  
                }  
            }  
        }  
    }  
}
```

```

        if(next == -1){
            pos = j;
            break;
        }

        if(next > farthest){
            farthest = next;
            pos = j;
        }

        frame[pos] = pages[i];
        faults++;
    }
}

printf("Page Hits = %d\n", hits);
printf("Page Faults = %d\n", faults);
printf("Hit Percentage = %.2f%%\n", (hits/12.0)*100);
printf("Fault Percentage = %.2f%%\n", (faults/12.0)*100);

return 0;
}

```

Output :

```

bhavi@LAPTOP-IH23AUG3 MSYS ~
$ ./opr
Page Hits = 5
Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%

```